2 The relational model: ER modelling

Week 2 · Week summary

Core ideas, lesson by lesson

1 Why the relational model won

- Hierarchical = tree of parents; network = graph of pointers. Both make programs navigate.
- Codd (1970): data as relations, linked by values, queried declaratively.
- It became practical in the early 1980s: faster hardware, query optimisers, and SQL.
- Data independence: change storage without rewriting programs.

2 Relations, tuples, and attributes

- Relation = table, tuple = row, attribute = column.
- Domain = the allowed values of an attribute.
- Degree = number of attributes; cardinality = number of tuples.
- No duplicate tuples, no meaningful order, one atomic value per cell.
- NULL marks a missing value; test it with IS NULL.

3 Superkeys, candidate keys, and primary keys

- Superkey: unique, possibly with extra attributes.
- Candidate key: a minimal superkey. There may be several.
- Primary key: the chosen candidate; unique, never NULL, minimal (irreducible), and stable. Alternate keys get UNIQUE.
- Composite key: unique only as a combination.
- Foreign key: a copy of another relation's primary key.

4 Foreign keys, NULL, and integrity rules

- Entity integrity: a primary key is never NULL.
- Referential integrity: a foreign key matches an existing key or is NULL.
- On delete: RESTRICT refuses, CASCADE propagates, SET NULL unlinks.
- NOT NULL on a foreign key makes the relationship mandatory.

5 1:1, 1:N, and N:M with keys and junction tables

- 1:N: foreign key on the "many" side.
- 1:1: foreign key on one side, plus UNIQUE.
- N:M: junction table with two foreign keys forming a composite primary key.
- Attributes of the pair go in the junction table.

6 ER modelling in depth

- Requirements → conceptual (ER) → logical (relations) → physical (SQL).
- Relationship degree: unary (recursive, with roles), binary, ternary.
- Cardinality = maximum; participation = minimum (total or partial).
- Weak entity: double rectangle, identifying relationship as a double diamond, partial key underlined with a dashed line.

7 Generalisation and specialisation

- Specialisation: top-down; generalisation: bottom-up. Both give an ISA hierarchy.
- Subtypes inherit all attributes and relationships of the supertype.
- Disjoint (d) vs overlapping (o); total (double line) vs partial (single line).
- Map with a table per entity, per subtype (total and disjoint only), or one table with a type column.

8 Chen vs crow's-foot notation

- Chen: rectangles, diamonds, ovals; 1/N on lines; double line = total.
- Crow's foot: boxes with attributes; | one, o zero, fork many.
- Inner mark = minimum, outer mark = maximum.
- Read each direction from the symbol at the far end.

9 Modelling a problem step by step

- Context → requirements → entities → attributes → relationships.
- Entities are nouns the organisation remembers many of, each identifiable.
- Totals such as monthly income are derived: compute them, do not store them.
- Tick every numbered requirement off against the final diagram.

10 From ER to a logical model, step by step

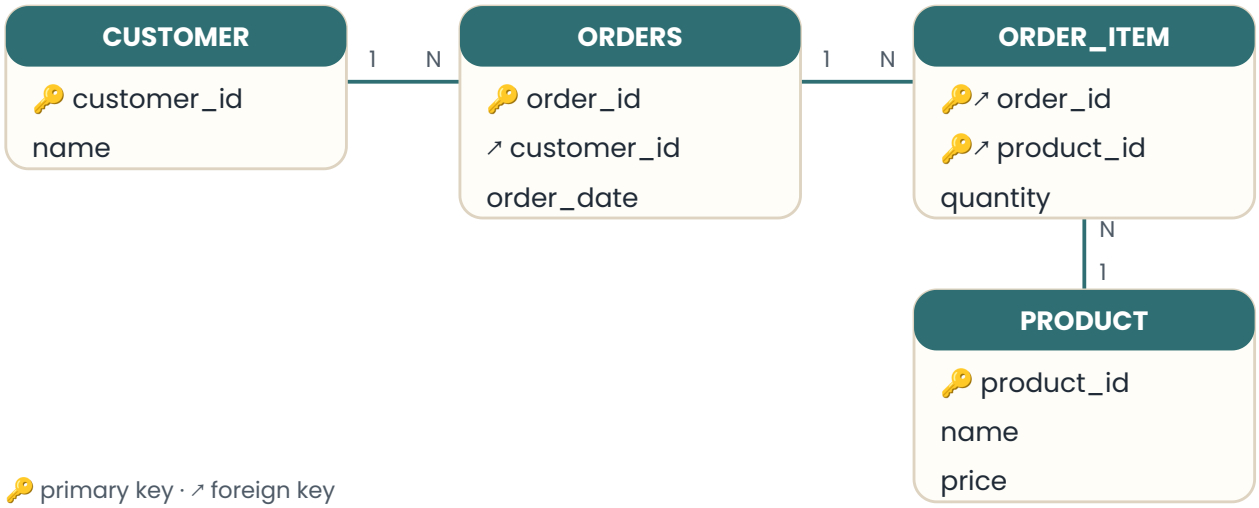
- Order: strong → weak → 1:1 → 1:N → N:M → multivalued → n-ary.
- Weak entity key = owner key + partial key.
- Total participation on the "many" side becomes NOT NULL.
- Check the result against the requirements, sentence by sentence.

11 The physical model: data types and DDL in MySQL and PostgreSQL

- Money: DECIMAL/NUMERIC(p,s), never FLOAT.
- Auto ids: AUTO_INCREMENT (MySQL) vs SERIAL/BIGSERIAL (PostgreSQL).
- DATETIME (MySQL) vs TIMESTAMP/TIMESTAMPTZ (PostgreSQL); BOOLEAN is TINYINT(1) in MySQL.
- Change a type: MODIFY (MySQL) vs ALTER COLUMN … TYPE (PostgreSQL).
- DELETE rows, TRUNCATE all rows, DROP the table.

12 Basic DML: INSERT, SELECT, UPDATE, DELETE

- INSERT with a column list; many rows in one statement; parents first.
- SELECT columns AS alias FROM t WHERE … ORDER BY … LIMIT n.
- UPDATE … SET a = 1, b = 2 WHERE …; DELETE FROM … WHERE ….
- Test the WHERE with a SELECT before changing data.



Watch out

Mistake	Fix
Putting the foreign key on the "one" side of a 1:N relationship.	The foreign key lives with the many: each song stores its album_id.
Forgetting UNIQUE on the foreign key of a 1:1 relationship.	Without UNIQUE, a 1:1 silently becomes 1:N.
Giving a weak entity only its partial key.	Its key is the owner's key plus the partial key: (pet_id, dose_no).
Thinking "relational" means "has relationships between tables".	The name comes from the mathematical relation: a table is a relation. Links between tables are a separate idea.
Believing older models vanished.	Some hierarchical systems still run in banks and airlines, and trees reappear in formats like XML and

	JSON documents.
Storing a list in one cell, such as genres = "pop, rock".	Cells are atomic. Put each genre in its own row of a separate table.
Writing WHERE album_id = NULL.	A comparison with NULL is never true. Write WHERE album_id IS NULL.
Confusing the cardinality of a relation with the cardinality of a relationship (1:N).	Same word, two ideas: row count of a table vs how many partners each entity can have.